

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456092.8812 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.023
Transit depth (Rp/Rs)^2	3.06 +/- 0.8 [%]
Orbital Inclination (inc)	86.09 +/- 0.78
Airmass coefficient 1 (a1)	1.002 +/- 0.013
Airmass coefficient 2 (a2)	-0.0001 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	1.28 %
Best Comparison Star	#3 - [50, 121]
Optimal Aperture	3.2
Optimal Annulus	5.28
Transit Duration (day)	0.0625 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.023 vs 0.1592 (deviation 0.66σ)
Duration fit vs published	0.0625 d vs 0.0946 d (deviation 3.68σ)
Mid-time O–C	3.7 min
Depth SNR	7.3
Residual scatter	1.28 %

Light curve

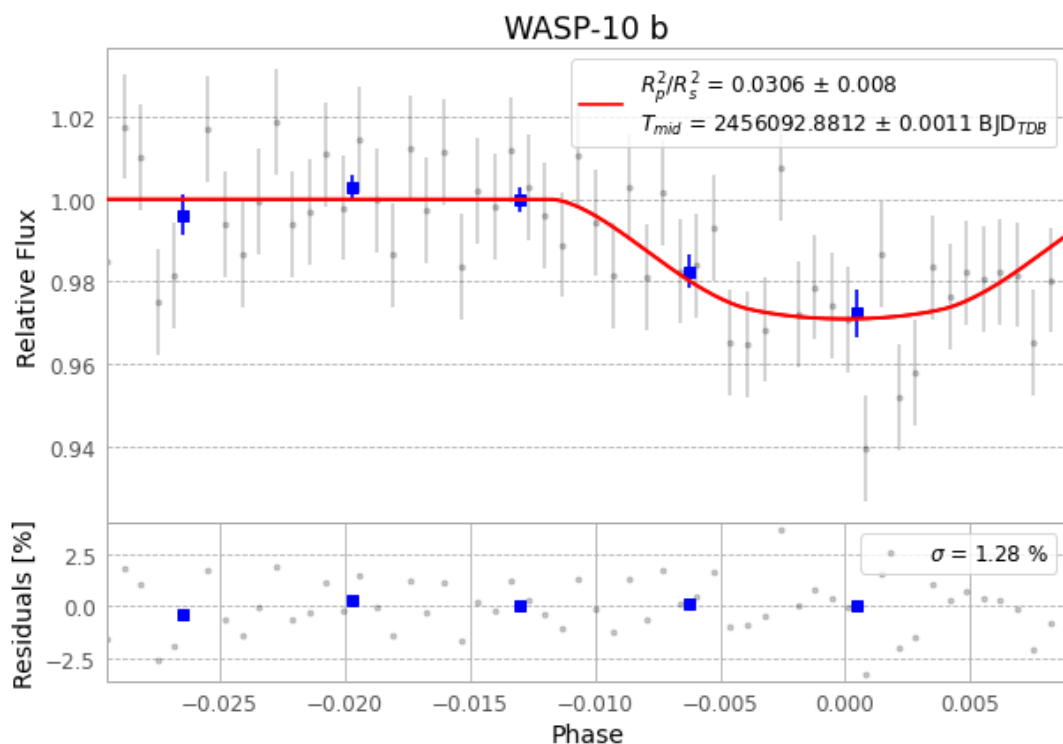


Figure 1 — FinalLightCurve_WASP-10 b_2012-06-13.png (SHA-256 8d40fbdf978785a5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [108, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 11285b001a460f3b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

58 FITS frames (38 MB), WASP-10120614065712.FITS → WASP-10120614094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-120614.log (f115f296b631...), FinalLightCurve_WASP-10 b_2012-06-13.png (8d40fbdf9787...), FinalLightCurve_WASP-10 b_2012-06-13.csv (39051ea56050...)

Compute resources

Wall time 102.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 00:48Z.